Year 7 Revision - Practice Answers

Answers

Each answer gives the result, with a short line of working where it helps.

A Operations with numbers

1 Y7 ICE-EM Ex 1L Q4

a $1 \times 5 = 5$ **d** $16 + 42 \div 6 = 16 + 7 = 23$ **f** $44 \div 2 \times 20 = 22 \times 20 = 440$
i $18 \div 3 + 8 \times 30 = 6 + 240 = 246$

2 Pack question

a $\dfrac{48 - 18}{5} = \dfrac{30}{5} = 6$ **b** $\dfrac{56 - 20}{12} = \dfrac{36}{12} = 3$

3 Y7 ICE-EM Ex 1H Q2

h $5646 \times 3 + 5646 \times 70 = 16\,938 + 395\,220 = 412\,158$

4 Pack question

a $587 = 6 \times 97 + 5$, so $587 \div 6 = 97\frac{5}{6}$
b $2270 = 12 \times 189 + 2$, so $2270 \div 12 = 189\frac{2}{12} = 189\frac{1}{6}$

5 Y7 ICE-EM Ex 2E Q6 One way for each; other mental methods are fine.

a $125 \times 32 = 125 \times 8 \times 4 = 1000 \times 4 = 4000$
b $135 \times 22 = 135 \times 20 + 135 \times 2 = 2700 + 270 = 2970$
d Halve four times, since $16 = 2 \times 2 \times 2 \times 2$: $1344 \to 672 \to 336 \to 168 \to 84$
e $367 \times 9 = 367 \times 10 - 367 = 3670 - 367 = 3303$

6 Y7 ICE-EM Ex 2I Q7

2, 3, 4, 5, 6, 8, 10. It ends in 0 (2, 5 and 10). Its last two digits 60 are divisible by 4. Its last three digits 760 are divisible by 8. Its digit sum is $5 + 8 + 4 + 7 + 6 + 0 = 30$, which is divisible by 3 but not by 9. It is even and divisible by 3, so it is divisible by 6.

7 Y7 ICE-EM Ex 2H Q3

a $224 = 2^5 \times 7$ and $336 = 2^4 \times 3 \times 7$.
HCF $= 2^4 \times 7 = 112$, LCM $= 2^5 \times 3 \times 7 = 672$

8 Y7 ICE-EM Ex 2H Q4

The LCM of 15 and 18 is 90, so it will be 90 minutes (1 hour 30 minutes), at 9:30 a.m.

9 Y7 Ch 2 Challenge Q10

$72 = 8 \times 9$. The last three digits $25\triangle$ must be divisible by 8, so they are 256 and $\triangle = 6$. The digit sum $\square + 7 + 2 + 5 + 6 = \square + 20$ must be a multiple of 9, so $\square = 7$. The number is $77\,256 = 72 \times 1073$.

B Computation with integers

1 Y7 ICE-EM Ex 11C Q3

c $15 - 1 = 14$ **f** $-28 + 35 + 2 = 9$ **j** $-92 + 54 + 82 = 44$ **l** $-500 + 2000 + 50 = 1550$

2 Y7 Ch 11 Review Q4

a $22 - (-3) = 25°\text{C}$ **b** $-3 - 22 = -25°\text{C}$

3 Y7 ICE-EM Ex 11D Q3

a 36 **d** $315 \times 20 = 6300$ **f** $102 \times (-2) = -204$ **i** $1 \times (-6) = -6$

4 Y7 ICE-EM Ex 11E Q6 Work from left to right.

a $8 \div 4 \times (-2) \times 3 = -12$ **b** $3 \times 4 \times (-5) \div 6 = -10$

5 Y7 ICE-EM Ex 11F Q9

b $-5 + 49 \div 49 + 2 = -5 + 1 + 2 = -2$ **d** $3 - 50 \div 25 - 2 = 3 - 2 - 2 = -1$

f $7 - 9 \times 4 = 7 - 36 = -29$ **h** $16 + 12 \div 4 - 4 = 16 + 3 - 4 = 15$

6 Y7 ICE-EM Ex 11F Q13

a Each person brings in $\$5 - \$2 = \$3$. $120 \times 3 - 500 = 360 - 500 = -140$: a $\$140$ loss.

b $120 \times (8 - 2) - 500 = 720 - 500 = \$220$ profit

c $60 \times 3 + 60 \times 6 - 500 = 180 + 360 - 500 = \$40$ profit

7 Y7 Ch 11 Challenge Q4 The first two numbers may be swapped.

a $4 + (-7) - (-2) = -1$ **b** $4 + (-2) - (-7) = 9$ **c** $-7 + (-2) - 4 = -13$

8 Y7 Ch 11 Challenge Q10

Group the terms in fours: $(100 + 99 - 98 - 97) + (96 + 95 - 94 - 93) + \cdots + (4 + 3 - 2 - 1)$. Each group equals 4 and there are 25 groups, so the sum is 100.

9 Y7 Ch 11 Challenge Q11

The five numbers add to $5 \times 2 = 10$. The other four add to $4 \times 4 = 16$. The smallest number is $10 - 16 = -6$.

C Indices

1 Y7 ICE-EM Ex 2D Q7

c $2^5 \times 3^5$ **f** $3^3 \times 7^5$

2 Y7 ICE-EM Ex 2D Q9

c $4 \times 25 = 100$ **d** $20^2 = 400$ **f** $4^3 = 64$ **g** $6 \times 10^5 = 600\,000$

3 Y7 ICE-EM Ex 2D Q6 To multiply powers of the same number, add the indices.

a 5 **b** 9 **d** 1 **f** 3

4 Y7 ICE-EM Ex 2F Q2

c $2^2 \times 3^2$ **d** $2^5 \times 3$ **f** 29^2 (841 is not divisible by any prime up to 23, and $29 \times 29 = 841$)

5 Y7 ICE-EM Ex 2G Q2

g $625 = 5^4 = (5^2)^2$, so the square root is 25

i $1444 = 2^2 \times 19^2$, so the square root is $2 \times 19 = 38$

j $5625 = 3^2 \times 5^4$, so the square root is $3 \times 5^2 = 75$

6 Y7 ICE-EM Ex 11F Q4

a 64 **b** -121 **d** $-9 \times 9 = -81$ **g** -125

7 Y7 ICE-EM Ex 11F Q5

c $4 \times (-64) = -256$ **g** $4 \times (-27) = -108$ **h** $7 \times (-1) = -7$

8 Y7 ICE-EM Ex 14C Q5

c $-0.027 + (-0.064) = -0.091$ **f** $-0.008 + 0.0016 = -0.0064$

9 Y7 Ch 2 Review Q14 Each pair 2×5 makes a factor of 10.

d $2^3 \times 5^4 = 10^3 \times 5 = 5000$: 3 zeros **f** 2×10^{20}: 20 zeros **g** 8×10^{20}: 20 zeros

10 Y7 Ch 2 Challenge Q11

a The factors are $2^0, 2^1, \ldots, 2^{20}$: 21 **b** $3^0, 3^1, \ldots, 3^6$: 7

c Each factor is $2^a \times 3^b$ with 21 choices for a and 7 choices for b: $21 \times 7 = 147$

D Fractions, decimals and percentages

1 Y7 ICE-EM Ex 4B Q6

q 7 **r** $\frac{171}{285} = \frac{9 \times 19}{15 \times 19} = \frac{3}{5}$ **t** $\frac{6 \times 7}{13 \times 7} = \frac{6}{13}$ **u** $\frac{3}{2}$

2 Y7 ICE-EM Ex 4E Q4

e $\frac{12}{15} + \frac{9}{15} = \frac{21}{15} = 1\frac{2}{5}$ **h** $\frac{20}{21} + \frac{15}{21} = \frac{35}{21} = 1\frac{2}{3}$ **k** $\frac{20}{30} + \frac{21}{30} = \frac{41}{30} = 1\frac{11}{30}$

3 Y7 Ch 5 Review Q2

a $\frac{2}{5} \times 4 = \frac{8}{5} = 1\frac{3}{5}$ **c** $18 + \frac{1}{2} = 18\frac{1}{2}$ **d** $\frac{4}{3} \times \frac{5}{4} + 2 = \frac{5}{3} + 2 = 3\frac{2}{3}$

4 Y7 ICE-EM Ex 5E Q2

e $\frac{5}{6} + \frac{4}{9} = \frac{15}{18} + \frac{8}{18} = \frac{23}{18} = 1\frac{5}{18}$

j $\frac{7}{8} + \frac{1}{3} - \frac{5}{12} = \frac{21}{24} + \frac{8}{24} - \frac{10}{24} = \frac{19}{24}$

o $\frac{2}{15} \times \frac{23}{24} \div \frac{7}{6} = \frac{2}{15} \times \frac{23}{24} \times \frac{6}{7} = \frac{23}{210}$

5 Y7 Ch 8 Review Q11

a $0.\dot{4}28571\dot{4}$ **b** $0.\dot{1}\dot{8}$ **d** $0.\dot{4}61538\dot{4}$ **f** $0.58\dot{3}$

6 Y7 Ch 14 Review Q4

f $-9.5 + (-7.77) = -17.27$ **i** $22.4 + 4 = 26.4$ **k** $18 \div (-12) = -1.5$

7 Y7 ICE-EM Ex 15B Q2

e $\frac{23}{50} \times 100\% = 46\%$ **i** $\frac{1250}{2000} \times 100\% = 62.5\%$ **l** $\frac{34.20}{60} \times 100\% = 57\%$

8 Y7 Ch 5 Review Q10

$\frac{2}{11} + \frac{3}{5} = \frac{10}{55} + \frac{33}{55} = \frac{43}{55}$, so the last 84 pages are $\frac{12}{55}$ of the book. Then $\frac{1}{55}$ of the book is 7 pages, and the book has 385 pages.

9 Y7 Ch 15 Challenge Q4

Let her money be $\$x$. After the dress she has $x - 24$. After the shirt she has 80% of that, so $\frac{4}{5}(x - 24) = \frac{2}{3}x$. Multiply by 15: $12x - 288 = 10x$, so $x = 144$. She had $\$144$.

10 Pack question

The new length is $\frac{4}{5}$ of the old length. To keep the area the same, the width must be multiplied by $\frac{5}{4} = 1.25$. The width must increase by 25%.

E Algebraic techniques

1 Y7 Ch 3 Review Q20

a $9a + b$ **c** $22a^2b$ **d** dc^2 is the same term as c^2d: $4c^2d + 5cd$

2 Y7 ICE-EM Ex 3F Q4

a $4x^2 \times x^3 = 4x^5$ **b** $3x \times 16x^2 = 48x^3$ **f** $30y^2z^3$

3 Pack question Divide or multiply before adding or subtracting.

a $9p + 6p = 15p$ **b** $6q - 4q = 2q$

4 Y7 ICE-EM Ex 16F Q2

f $6x + 18 + 5x - 30 = 11x - 12$ **g** $8x + 12 + 5x - 20 = 13x - 8$
h $12x + 6 + 7x + 28 = 19x + 34$

5 Pack question

a $9y^2 + 6y^2 - 21y = 15y^2 - 21y$ **b** $4p^2 + 12p - 5p = 4p^2 + 7p$

6 Y7 ICE-EM Ex 7B Q1

d $\frac{5x - 2}{5}$ **j** $4\left(\frac{2x}{5} - 3\right)$ **k** $\frac{3}{7}\left(23 - \frac{4x}{9}\right)$

7 Y7 ICE-EM Ex 12A Q10

a i $\$(1000 - x)$ ii $\$(1000 - 5x)$

b i $1000 - 500 = \$500$ ii $\$0$ iii $1000 - 1250 = -\$250$ (the account is overdrawn by $\$250$)

8 Y7 ICE-EM Ex 7B Q10

a $3(12 + 10) = 66$ **c** $6 + \frac{18-17}{4} = 6\frac{1}{4}$ **d** $29 \div \frac{1}{2} - 24 = 58 - 24 = 34$

9 Y7 ICE-EM Ex 12A Q9

d $4^3 = 64$ **e** $2^3 + 4 = 12$ **f** $-8 - 40 = -48$

10 Y7 Ch 7 Challenge Q7

a $BC = \dfrac{16}{x}$ cm **b** Perimeter $= 2x + \dfrac{32}{x}$ cm

c $8 + 8 = 16$ cm **d** $\frac{1}{2} + 128 = 128\frac{1}{2}$ cm

F Equations

1 Y7 ICE-EM Ex 16C Q3

d $4x = 12, x = 3$ **j** $\frac{x}{12} = 17, x = 204$ **l** $\frac{z}{7} = 12, z = 84$

2 Y7 Ch 16 Review Q2

e $5x = 48, x = 9\frac{3}{5}$ **k** $x = -20$ **p** $15x - 25 = 17, 15x = 42, x = 2\frac{4}{5}$

3 Pack question

a $x + 7 = 24, x = 17$ **b** $2y - 1 = 15, 2y = 16, y = 8$

4 Y7 Ch 16 Review Q6

g $-3x = -16, x = 5\frac{1}{3}$ **h** $4x = -9, x = -2\frac{1}{4}$ **i** $-2x = -22, x = 11$

5 Y7 ICE-EM Ex 16E Q3

d $6x - 42 = 12, 6x = 54, x = 9$ **f** $28n - 70 = 11, 28n = 81, n = 2\frac{25}{28}$

6 Y7 ICE-EM Ex 16G Q3

b $60 + 6x = 4x + 100, 2x = 40, x = 20$ **d** $60 = 3x + 45, 3x = 15, x = 5$

7 Pack question

a Area $= 4(3x + 2) = 12x + 8$ cm^2

b $12x + 8 = 50, 12x = 42, x = 3.5$

8 Y7 Ch 16 Review Q14

Let Brett receive $\$x$. Then Jennifer receives $\$(x + 16)$ and David receives $\$(x + 24)$.

So $3x + 40 = 400, x = 120$.

Brett $\$120$, Jennifer $\$136$, David $\$144$

9 Y7 Ch 16 Challenge Q6

Let Renae have $\$r$ at first, so James has $\$(60 - r)$. Then $r - 12 = 2(60 - r + 12)$, so $r - 12 = 144 - 2r$, $3r = 156$ and $r = 52$. Renae had $\$52$.

10 Pack question

Adding the three totals counts each box twice: $2(A + B + C) = 27 + 33 + 30 = 90$, so $A + B + C = 45$.
$C = 45 - 27 =$ 18 kg, $A = 45 - 33 =$ 12 kg, $B = 45 - 30 =$ 15 kg

G Linear relationships

1 Y7 Ch 8 Challenge Q1

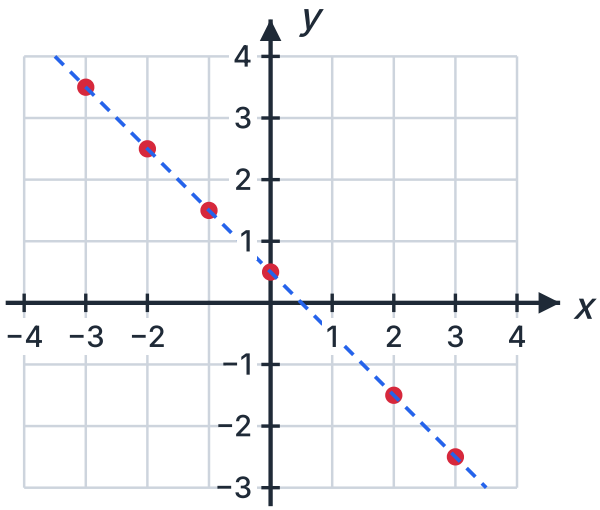
c Subtract 0.1 each time: 12.2, 12.1, 12.0, 11.9, 11.8
d Subtract 1.1 each time: 9.1, 8.0, 6.9, 5.8, 4.7

2 Y7 Ch 12 Review Q3

$A(4, 4)$, $B(-4, 3)$, $C(-3, -2)$, $D(3, -3)$, $E(1, 1)$, $F(0, -4)$, $G(-2, 0)$

3 Y7 ICE-EM Ex 12C Q2

d For $x = -3, -2, -1, 0, 1, 2, 3$ the values of y are
$3\frac{1}{2}$, $2\frac{1}{2}$, $1\frac{1}{2}$, $\frac{1}{2}$, $-\frac{1}{2}$, $-1\frac{1}{2}$, $-2\frac{1}{2}$.
The points are $(-3, 3\frac{1}{2})$, $(-2, 2\frac{1}{2})$, $(-1, 1\frac{1}{2})$, $(0, \frac{1}{2})$, $(1, -\frac{1}{2})$, $(2, -1\frac{1}{2})$, $(3, -2\frac{1}{2})$. They lie on a straight line.



4 Y7 ICE-EM Ex 12D Q2

a For 4, 5 and 6 triangles: 9, 11, 13 matches **b** 2 matches
c The points $(1, 3)$, $(2, 5)$, $(3, 7)$, $(4, 9)$, $(5, 11)$, $(6, 13)$ lie on a straight line. **d** $m = 2t + 1$

5 Y7 Ch 12 Review Q10

c $n = 4 \times m - 4$ **d** $q = 2 \times p - 1$ **f** $y = -3 \times x - 7$

6 Y7 ICE-EM Ex 12D Q8

a $y = x + 3$ **c** $y = -2x + 1$ **f** $y = -2x - 1$

7 Y7 ICE-EM Ex 12D Q6

a For months 2 to 6: 960, 940, 920, 900, 880 **b** $d = 1000 - 20m$

c $1000 - 200 = \$800$ **d** $1000 \div 20 = 50$ months

8 Y7 Ch 12 Challenge Q4

The base AB is 5 cm, so $\frac{1}{2} \times 5 \times \text{height} = 20$ and the height is 8 cm. C can be any point with y-coordinate 8 or -8, that is, any point on the line $y = 8$ or the line $y = -8$.

9 Pack question

a $2 = 2 \times 3 + c$, so $c = -4$.

b When $y = 0$, $2x - 4 = 0$ and $x = 2$. It crosses the x-axis at $(2, 0)$ and the y-axis at $(0, -4)$.

c $\frac{1}{2} \times 2 \times 4 = 4$ square units

H Probability

1 Y7 ICE-EM Ex 17A Q1

c $\frac{1}{2}$ and $\frac{1}{2}$ **d** 1 **e** 0

2 Y7 ICE-EM Ex 17B Q1

a 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 **b** 1, 3, 5, 7, 9

3 Y7 ICE-EM Ex 17B Q12 There are 30 sweets.

b $\frac{9+6}{30} = \frac{1}{2}$ **c** $\frac{20}{30} = \frac{2}{3}$ **e** $\frac{10+9}{30} = \frac{19}{30}$

4 Pack question

a 3, 9, 15: $\frac{3}{20}$ **b** 10 odd numbers and 6 multiples of 3, with 3 counted twice: $\frac{10+6-3}{20} = \frac{13}{20}$

c 15, 16, 17, 18, 19, 20: $\frac{6}{20} = \frac{3}{10}$

5 Y7 ICE-EM Ex 17B Q14 There are $250 + 400 + 175 + 75 = 900$ jelly beans.

a $\frac{250}{900} = \frac{5}{18}$ **b** $\frac{425}{900} = \frac{17}{36}$ **c** $\frac{250}{900} = \frac{5}{18}$ **d** $\frac{75}{900} = \frac{1}{12}$ **e** $\frac{475}{900} = \frac{19}{36}$

6 Pack question

a Plays tennis: swims 9, does not swim 14, total 23.

Does not play tennis: swims 22, does not swim 15, total 37.

Total: swims 31, does not swim 29, total 60.

b i $\frac{22}{60} = \frac{11}{30}$ ii $\frac{15}{60} = \frac{1}{4}$

c If k students are added, $\frac{23+k}{60+k} = \frac{1}{2}$, so $46 + 2k = 60 + k$ and $k = 14$.

7 Y7 Ch 17 Challenge Q5

If r red balls are left, $\frac{r}{r+8} = \frac{1}{3}$, so $3r = r + 8$ and $r = 4$. So 11 red balls were removed.

8 Y7 Ch 17 Challenge Q1

f 7, 17, 27, 37, 47: $\frac{1}{10}$ **h** 2, 12, 20 to 29, 32, 42 (14 tickets): $\frac{7}{25}$
i the same 14 tickets except 22: $\frac{13}{50}$ **j** $50 - 14 = 36$ tickets: $\frac{18}{25}$

9 Pack question

Let $P(\text{blue}) = P(\text{green}) = b$. Then $\frac{1}{3} + b + b + 2b = 1$, so $4b = \frac{2}{3}$ and $b = \frac{1}{6}$. So $P(\text{yellow}) = \frac{1}{3}$, and the angle is $\frac{1}{3} \times 360° = 120°$.

I Mixed challenge

1 Y7 ICE-EM Ex 10A Ch 1 Q16

a $20 + 3 - 1 = 22$ **b** $50 \div 2 - 1 = 24$ **c** $20 + 3 = 23$

2 Y7 ICE-EM Ex 22A Ch 15 Q4 Write each as a decimal first.

a $4\frac{1}{2}\% = 0.045$ and $\frac{12}{32} = 0.375$: $4\frac{1}{2}\%,\ \frac{12}{32},\ 0.39$
b $64\% = 0.64$ and $\frac{2}{3} = 0.\dot{6}$: $0.6,\ 64\%,\ \frac{2}{3}$

3 Y7 ICE-EM Ex 10A Ch 5 Q6

a $\frac{17}{12} \div \frac{1}{6} = \frac{17}{12} \times 6 = 8\frac{1}{2}$ **b** $\frac{1}{6} \div \frac{5}{6} = \frac{1}{5}$ **c** $\frac{2}{3} \div \frac{1}{2} = 1\frac{1}{3}$

4 Y7 ICE-EM Ex 10A Ch 5 Q9

$\frac{2}{3} \times \frac{3}{4} \times \frac{4}{5} = \frac{2}{5}$ of the money is left.

5 Y7 ICE-EM Ex 10A Ch 5 Q10

The bells ring every 1, $\frac{5}{4}$, $\frac{6}{5}$ and $\frac{7}{6}$ seconds. A time t seconds works when t, $\frac{4t}{5}$, $\frac{5t}{6}$ and $\frac{6t}{7}$ are all whole numbers, so t is a multiple of 5, 6 and 7. The smallest is 210 seconds (3 minutes 30 seconds).

6 Y7 ICE-EM Ex 22A Ch 16 Q9

$x + 13 = 4 - \frac{x}{2}$, so $\frac{3x}{2} = -9$ and $x = -6$.

7 Y7 ICE-EM Ex 22A Ch 17 Q2

d 15, 17, 19, 21, 23: $\frac{5}{24}$ **e** $\frac{18}{24} = \frac{3}{4}$ **g** 5, 10, 15, 20: $\frac{1}{6}$ **h** 2, 3, 5, 7, 11, 13, 17, 19, 23: $\frac{3}{8}$

8 Y7 ICE-EM Ex 10B Evens and odds An odd sum means one number is even and the other is odd.

1 sometimes true and sometimes false ($3 + 6$: $6 \div 3 = 2$) **2** always true **3** always false
4 sometimes true and sometimes false ($1 + 2$: $2 < 3$; $3 + 4$: $12 > 7$)
5 sometimes true and sometimes false (adding 1 to the odd number gives an even product; adding 1 to the even number gives two odd numbers)